Department of Computer Science and Engineering

KPR Institute of Engineering and Technology

B.E. – Computer Science and Engineering

Laboratory Record

U21CS504

Web Technologies Laboratory

Regulation 2021

Nov / Dec 2026

Department of Computer Science and Engineering

KPR Institute of Engineering and Technology

Name : _____

Roll Number : _____

Subject Code & Title : _____

Department : _____

Year & Semester : _____

This is the certified record of work done

by…… Register Number

……

…….

Faculty In- Charge **Head of the Department**

Place:

Date:

He/ She has submitted the record for the End Semester Practical Examination held on ……

Internal Examiner External Examiner

List of Experiments

Expr. No.	Name of the Experiments	Page No.	Marks (20)	Faculty Sign
1.	Static web page design using HTML and CSS			
2.	BMI Calculator using JavaScript			
3.	User registration form validation using JavaScript			
4.	Implementing web applications with API calls using Java Script			
5.	Event registration form using PHP and MySQL			
6.	Web application for food delivery system using PHP and MySQL			
7.	Web application for leave management using Servlet and MySQL			
8.	Web application for online shopping using Servlet and MySQL			
9.	Mini Project:			

Rubrics for Laboratory Experiments

Criteria & Points / Content	Strongly Acceptable	Acceptable	Considerable	Below Expectations	Unacceptable
	4	3	2	1	0
Understanding the Problem Statement	The procedure is explained and well designed the problem with appropriate analysis	Some programs are written accurately	Major errors in program writing	Programs are not written/wrong program written	Programs are not written/wrong program written
Completion of target in Lab	100% target has been completed	80% target has been completed	70% target has been completed	50% target has been completed	25% target has been completed
User Interface Design	Well designed problem with appropriate analysis	Designed the problem with analysis	Missing some experimental procedure with partial analysis	Missing major experimental details and analysis	Could not be able to explain the program
Development	Implement Optimal solution with appropriate results	80% of implementation has been done	65% of implementation has been done. Few procedures were explained	No proper explanation on implementation	Could not be able to explain the program
Innovation	Most of the innovations are applied clearly	Some of the innovations are applied undoubtedly	Few innovations are applied clearly	innovations applied	None of the innovations used

Ex No: 01	Static web page design using HTML and CSS

AIM:

To design and develop a responsive Login and Sign Up web page using HTML and CSS with a user-friendly interface, styled input fields, buttons, and attractive page layout.

PROCEDURE:

1. Create a project folder named Login Page.
2. Create the following files:
   - index.html
   - style.css
3. Create the basic HTML structure using DOCTYPE, html, head, and body tags.
4. Design the login/sign-up form using:
   - Heading
   - Labels
   - Text input
   - Email input
   - Password input
   - Confirm Password input
   - Submit button
5. Link the CSS file with the HTML file using the <link> tag.
6. Apply CSS styling to improve the appearance of the webpage.
7. Add background colors, border radius, padding, margins, shadows, and button effects.
8. Open the index.html file using a web browser or Live Server.
9. Verify that all input fields and buttons are displayed properly.

HTML USED:

- <!DOCTYPE html>
- <html>
- <head>
- <meta>
- <title>
- <link>
- <body>
-
- <h1>

- <form>
- <label>
- <input>
- <button>

CSS USED:

- background-color
- font-family
- font-size
- font-weight
- color
- width
- height
- padding
- margin
- border
- border-radius
- box-shadow
- display
- justify-content
- align-items
- text-align
- cursor
- hover
- transition

INDEX.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Sign Up</title>

    <link rel="stylesheet" href="style.css">
</head>
<body>

    <div class="signup-container">

        <!-- User Icon -->
        <div class="user-icon">
            <div class="head"></div>
            <div class="body"></div>
        </div>

        <h1>Sign Up</h1>

        <form>
```

```
            <label for="name">Name</label>
            <input type="text" id="name" placeholder="Enter your name">

            <label for="email">Email</label>
            <input type="email" id="email" placeholder="Enter your email">

            <label for="password">Password</label>
            <input type="password" id="password" placeholder="Enter your password">

            <label for="confirm-password">Confirm Password</label>
            <input type="password" id="confirm-password" placeholder="Confirm your password">

            <button type="submit">SignUp</button>

        </form>

    </div>

</body>
</html>
```

STYLE.CSS

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    min-height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    background-color: white;
    font-family: Arial, sans-serif;
}

.signup-container {
    width: 625px;
    min-height: 750px;
    background-color: #e98bd1;
    border-radius: 12px;
    padding: 40px 63px;
    box-shadow: 8px 8px 15px rgba(0, 0, 0, 0.35);
}

/* User Icon */
.user-icon {
    width: 100px;
    height: 100px;
    background-color: white;
    border-radius: 50%;
    margin: 0 auto 17px;
```

```
   border: 8px solid white;
    outline: 4px solid #111;

    position: relative;
}

/* Head */
.head {
    position: absolute;
    width: 30px;
    height: 30px;
    background-color: #111;
    border-radius: 50%;
    top: 14px;
    left: 27px;
}

/* Body */
.body {
    position: absolute;
    width: 58px;
    height: 35px;
    background-color: #111;
    border-radius: 35px 35px 0 0;
    bottom: 10px;
    left: 13px;
}

h1 {
    text-align: center;
    font-size: 32px;
    color: #000;
    margin-bottom: 38px;
}

form {
    width: 100%;
}

label {
    display: block;
    font-size: 20px;
    font-weight: bold;
    margin: 0 0 25px 13px;
}

input {
    width: 100%;
    height: 58px;
    border: none;
    border-radius: 7px;
    padding: 0 11px;
    margin-bottom: 25px;
```

```
    background-color: #eef8ff;
    font-size: 16px;

    box-shadow: 3px 4px 5px rgba(0, 0, 0, 0.3);
}

input::placeholder {
    color: #777;
}

input:focus {
    outline: 2px solid #42c5e8;
}

button {
    display: block;
    width: 250px;
    height: 52px;

    margin: -1px auto 0;

    background-color: #48c6e5;
    border: none;
    border-radius: 6px;

    font-size: 16px;
    font-weight: bold;
    color: black;

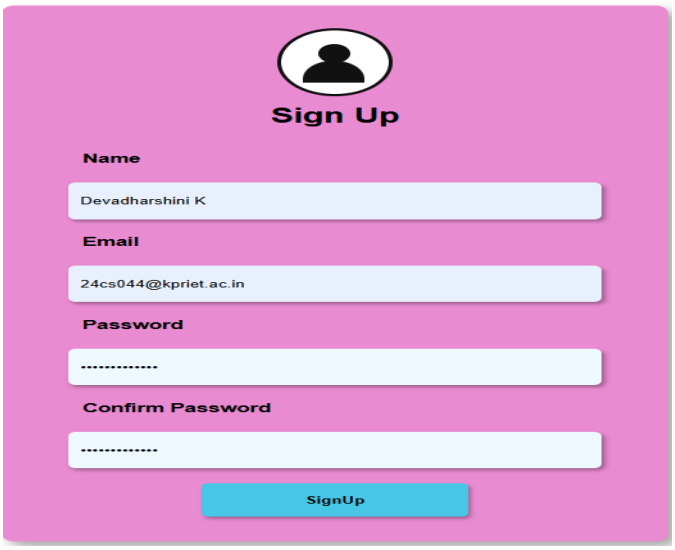
    box-shadow: 3px 4px 5px rgba(0, 0, 0, 0.3);

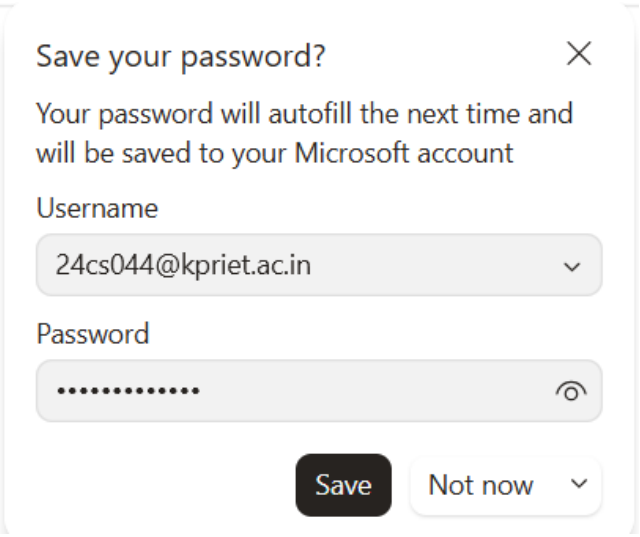
    cursor: pointer;
}

button:hover {
    background-color: #32b9dc;
}

button:active {
    transform: translateY(2px);
}
```

OUTPUT:





RUBRICS:

Rubrics	Actual Marks	Awarded Marks
Understanding (R1)	4	
Timeline (R2)	4	
Design (R3)	4	
Development (R4)	4	
Innovation (R5)	4	
Total Marks	**20**	

RESULT:

A basic multi-page company website was successfully designed and developed using HTML and CSS. The website contains Home, About, Products/Services, and Contact pages connected through a common navigation menu. CSS styling was applied to improve the appearance, and all hyperlinks, images, tables, and forms function correctly.

Ex No: 02	BMI Calculator using JavaScript

AIM:

To design and develop a BMI (Body Mass Index) Calculator using HTML, CSS, and JavaScript that calculates and displays the BMI based on the user's height and weight.

PROCEDURE:

1. Create a project folder named BMI Calculator.
2. Create the following files:
    - index.html
    - style.css
    - script.js
3. Design the BMI Calculator interface using HTML.
4. Create input fields for:
    - Weight in kilograms
    - Height in centimeters
5. Create Calculate BMI and Reset buttons.
6. Apply CSS to create an attractive and user-friendly interface.
7. Write JavaScript code to read the height and weight values.
8. Convert height from centimeters to meters.
9. Calculate BMI using the formula:

BMI = Weight (kg) / Height² (m)

10. Display the calculated BMI value and its corresponding category.
11. Validate the input values and display an appropriate message for invalid input.
12. Test the application by entering different height and weight values.

HTML USED:

- <!DOCTYPE html>
- <html>
- <head>
- <meta>
- <title>
- <link>
- <body>
-
- <h2>
- <label>
- <input>
- <button>
- <script>

CSS PROPERTIES:

- background
- color
- font-family
- width
- height
- padding
- margin
- border
- border-radius
- text-align
- display
- justify-content
- align-items
- box-shadow
- cursor
- hover

JAVASCRIPT CONCEPTS USED

- Variables
- Functions
- parseFloat()
- getElementById()
- Conditional statements
- if
- else if
- else
- Arithmetic operators
- toFixed()
- DOM manipulation
- Button click event

INDEX.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>BMI Calculator</title>

    <link rel="stylesheet" href="style.css">
</head>

<body>

    <div class="container">
```

```
<div class="card">

<div class="icon">

</div>

<h1>BMI Calculator</h1>

<p class="subtitle">
  Check your Body Mass Index
</p>

<div class="input-group">
  <label for="weight">Weight</label>

  <div class="input-box">
    <input type="number" id="weight" placeholder="Enter your weight">
    <span>kg</span>
  </div>
</div>

<div class="input-group">
  <label for="height">Height</label>

  <div class="input-box">
    <input type="number" id="height" placeholder="Enter your height">
    <span>cm</span>
  </div>
</div>

<button onclick="calculateBMI()">
  Calculate BMI
</button>

<button class="reset-btn" onclick="resetBMI()">
  Reset
</button>

<div class="result" id="result">

  <h2>Your BMI</h2>

  <div class="bmi-value" id="bmiValue">
    --
  </div>

  <p id="category">
    Enter your details to calculate
  </p>

</div>

<div class="info">
```

```
                <div>
                <span class="dot underweight"></span>
                Underweight


                <b>&lt; 18.5</b>
            </div>

            <div>
                <span class="dot normal"></span>
                Normal
                <b>18.5 - 24.9</b>
            </div>

            <div>
                <span class="dot overweight"></span>
                Overweight
                <b>25 - 29.9</b>
            </div>

            <div>
                <span class="dot obese"></span>
                Obese
                <b>30+</b>
            </div>

        </div>

    </div>

    <script src="script.js"></script>

</body>
</html>
```

STYLE.CSS:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>BMI Calculator</title>

    <link rel="stylesheet" href="style.css">
</head>

<body>

    <div class="container">

        <div class="card">
```

```
<div class="icon">

</div>

<h1>BMI Calculator</h1>

<p class="subtitle">
   Check your Body Mass Index
</p>

<div class="input-group">
   <label for="weight">Weight</label>

   <div class="input-box">
      <input type="number" id="weight" placeholder="Enter your weight">
      <span>kg</span>
   </div>
</div>

<div class="input-group">
   <label for="height">Height</label>

   <div class="input-box">
      <input type="number" id="height" placeholder="Enter your height">
      <span>cm</span>
   </div>
</div>

<button onclick="calculateBMI()">
   Calculate BMI
</button>

<button class="reset-btn" onclick="resetBMI()">
   Reset
</button>

<div class="result" id="result">

   <h2>Your BMI</h2>

   <div class="bmi-value" id="bmiValue">
      --
   </div>

   <p id="category">
      Enter your details to calculate
   </p>

</div>

<div class="info">

   <div>
      <span class="dot underweight"></span>
      Underweight
```

```
                    <b>&lt; 18.5</b>

                </div>
                <div>
                    <span class="dot normal"></span>
                    Normal
                    <b>18.5 - 24.9</b>
                </div>


                    <div>
                    <span class="dot overweight"></span>
                    Overweight
                    <b>25 - 29.9</b>
                </div>

                <div>
                    <span class="dot obese"></span>
                    Obese
                    <b>30+</b>
                </div>

            </div>

        </div>

    </div>

    <script src="script.js"></script>

</body>
</html>
```

SCRIPT JS:

```
function calculateBMI() {

    let weight = parseFloat(
        document.getElementById("weight").value
    );

    let height = parseFloat(
        document.getElementById("height").value
    );

    if (isNaN(weight) || isNaN(height) ||
        weight <= 0 || height <= 0) {

        alert("Please enter valid weight and height!");

        return;
    }
```

```
    // Convert height from cm to meters

    let heightInMeter = height / 100;

    // BMI Formula

    let bmi = weight / (heightInMeter * heightInMeter);

    bmi = bmi.toFixed(1);

    document.getElementById("bmiValue").innerText = bmi;

    let category = "";

    if (bmi < 18.5) {

        category = "Underweight";

    }
    else if (bmi >= 18.5 && bmi < 25) {

        category = "Normal Weight";

    }
    else if (bmi >= 25 && bmi < 30) {

        category = "Overweight";

    }
    else {

        category = "Obese";

    }

    document.getElementById("category").innerText = category;
}

function resetBMI() {

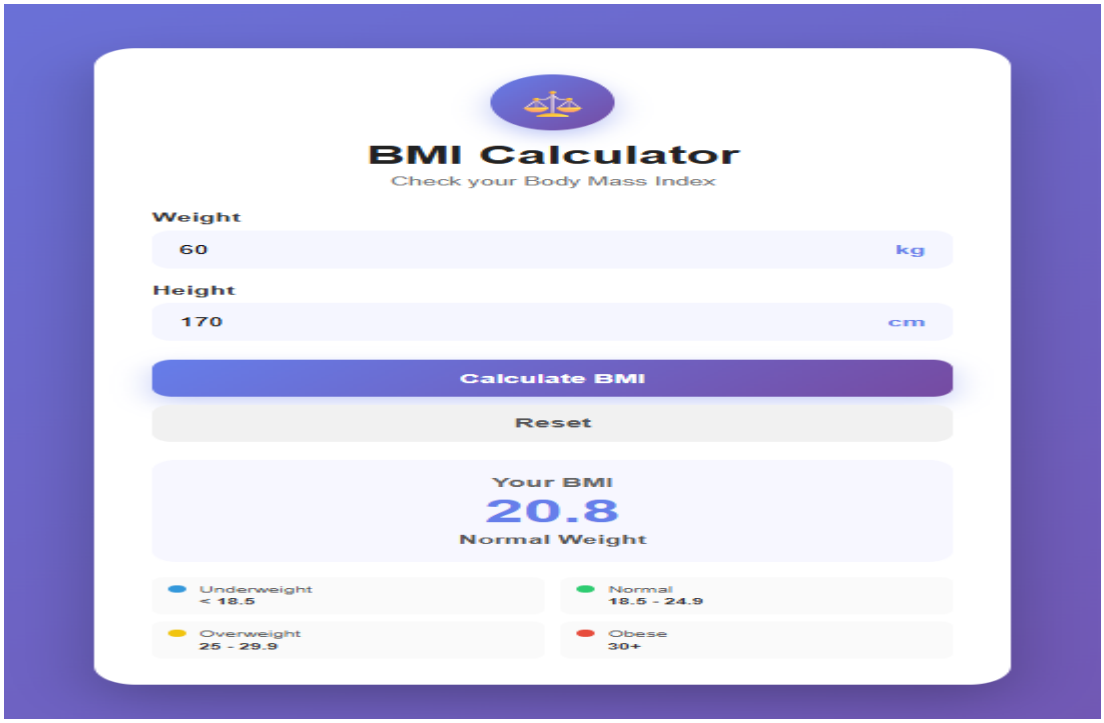
    document.getElementById("weight").value = "";

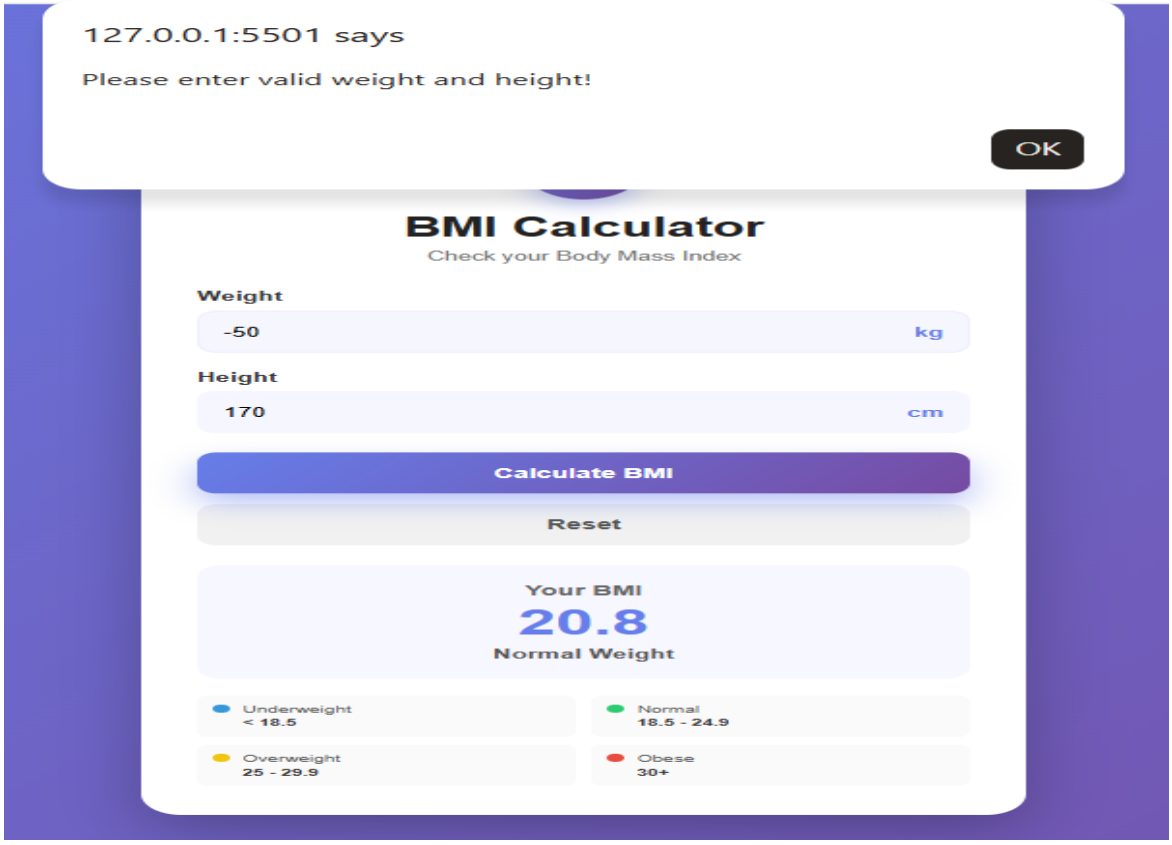
    document.getElementById("height").value = "";

    document.getElementById("bmiValue").innerText = "--";

    document.getElementById("category").innerText =
        "Enter your details to calculate";
}
```

OUTPUT:


BMI Calculator
Check your Body Mass Index
Weight
60
kg
Height
170
cm
Calculate BMI
Reset
Your BMI
20.8
Normal Weight
Underweight
< 18.5
Normal
18.5 - 24.9
Overweight
25 - 29.9
Obese
30+


127.0.0.1:5501 says
Please enter valid weight and height!
OK
BMI Calculator
Check your Body Mass Index
Weight
-50
kg
Height
170
cm
Calculate BMI
Reset
Your BMI
20.8
Normal Weight
Underweight
< 18.5
Normal
18.5 - 24.9
Overweight
25 - 29.9
Obese
30+

RUBRICS:

Rubrics	Actual Marks	Awarded Marks
Understanding (R1)	4	
Timeline (R2)	4	
Design (R3)	4	
Development (R4)	4	
Innovation (R5)	4	
Total Marks	**20**	

RESULT:

A BMI Calculator was successfully designed and developed using HTML, CSS, and JavaScript. The application accepts the user's height and weight, accurately calculates the BMI, and displays the corresponding health category.

Ex No: 03	Develop a User Registration Form with Client-Side Validation using JavaScript

AIM:

To develop a user registration form using HTML and CSS and perform client-side form validation using JavaScript with regular expressions, DOM manipulation, and event handling.

PROCEDURE:

1. Create a project folder named User_Registration_Validation.
2. Create three files named index.html, style.css, and validation.js.
3. Create the HTML structure and design a registration form with fields for name, email, mobile number, password, confirm password, gender, date of birth, city, and terms and conditions.
4. Add Register and Reset buttons and separate areas for displaying validation error messages.
5. Apply CSS styling to the registration form using suitable colors, fonts, spacing, borders, border radius, shadows, and button hover effects.
6. Use JavaScript to read the values entered by the user using DOM methods and remove unnecessary spaces using trim().
7. Validate the name, email, mobile number, password, confirm password, gender, date of birth, city, and terms and conditions.
8. Use Regular Expressions to validate the name, email, mobile number, and password fields.
9. Display appropriate error messages for invalid inputs and prevent form submission using event.preventDefault().
10. Display a registration success message with the user's name when all validations are successfully completed.

HTML USED:

- <!DOCTYPE html>
- <html>
- <head>
- <title>
- <body>
-
- <h2>
- <form>
- <label>
- <input>
-
- <select>
- <option>
- <button>

CSS PROPERTIES USED:

- margin
- padding
- background
- background-color
- font-family
- text-align
- width
- max-width
- border
- border-radius
- box-shadow
- display
- color
- font-size
- font-weight
- cursor
- box-sizing

JAVASCRIPT CONCEPTS USED:

- Variables
- Functions
- Conditional Statements
- Regular Expressions (RegEx)
- Event Handling
- DOM Manipulation
- document.getElementById()
- document.querySelector()
- value
- trim()
- textContent
- innerHTML
- event.preventDefault()
- classList.add()
- classList.remove()
- Form Validation
- Form Reset

INDEX.HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>User Registration Form Validation</title>
<!-- External CSS -->
<link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
<!-- Header -->
<div class="header">
<div class="brand-box">F</div>
<div>
<p class="small-title">WEB TECHNOLOGIES LAB</p>
<h1>Registration Portal</h1>
</div>
<div class="status" id="statusText">
INCOMPLETE
</div>
</div>
<div class="intro">
<p>
Create your account by entering the required information.
All fields are validated using JavaScript.
</p>
</div>
<!-- Validation Progress -->
<div class="progress-section">
<div class="progress-info">
<span>Form Completion</span>
<span id="progressText">0%</span>
</div>
<div class="progress-bar">
<div id="progressFill"></div>
</div>
</div>
<!-- Task 3: Main Registration Container -->
<div class="form-card">
<h2>User Registration Form</h2>
<form id="registrationForm" novalidate>
<!-- Full Name -->
<div class="form-group">
<label for="fullName">Full Name</label>
<input
type="text"
id="fullName"
name="fullName"
placeholder="Enter your full name"
required
```

```
minlength="3"
autocomplete="name">
<span class="error" id="nameError"></span>
</div>
<!-- Email -->
<div class="form-group">
<label for="email">Email Address</label>
<input
type="email"
id="email"
name="email"
placeholder="example@gmail.com"
required
autocomplete="email">
<span class="error" id="emailError"></span>
</div>
<!-- Mobile Number -->
<div class="form-group">
<label for="mobile">Mobile Number</label>
<input
type="text"
id="mobile"
name="mobile"
placeholder="10-digit mobile number"
required
maxlength="10"
inputmode="numeric"
autocomplete="tel”  >
<span class="error" id="mobileError"></span>
</div>
<!-- Password -->
<div class="form-group">
<label for="password">Password</label>
<input
type="password"
id="password"
name="password"
placeholder="Create a strong password"
required
minlength="8"
autocomplete="new-password">
<div class="password-info">
<span id="passwordCount">0 characters</span>
<span id="passwordStrength">Strength: -</span>
</div>
<span class="error" id="passwordError"></span>
</div>
<!-- Confirm Password -->
<div class="form-group">
<label for="confirmPassword">Confirm Password</label>
<input
type="password"
```

```
id="confirmPassword"
name="confirmPassword"
placeholder="Re-enter your password"
required
autocomplete="new-password">
<span class="error" id="confirmPasswordError"></span>
</div>
<!-- Gender -->
<div class="form-group">
<label>Gender</label>
<div class="radio-group">
<label class="radio-option">
<input
type="radio"
name="gender"
value="Male"
required>
Male
</label>
<label class="radio-option">
<input
type="radio"
name="gender"
value="Female">
Female
</label>
<label class="radio-option">
<input
type="radio"
name="gender"
value="Other">
Other
</label>
</div>
<span class="error" id="genderError"></span>
</div>
<!-- Date of Birth -->
<div class="form-group">
<label for="dob">Date of Birth</label>
<input
type="date"
id="dob"
name="dob"
required>
<span class="error" id="dobError"></span>
</div>
<!-- City -->
<div class="form-group">
<label for="city">City</label>
<select id="city" name="city" required>
<option value="">-- Select Your City --</option>
```

```
<option value="Coimbatore">Coimbatore</option>
<option value="Chennai">Chennai</option>
<option value="Tirupur">Tirupur</option>
<option value="Erode">Erode</option>
<option value="Salem">Salem</option>
<option value="Madurai">Madurai</option>
<option value="Trichy">Trichy</option>
<option value="Bangalore">Bangalore</option>
<option value="Kochi">Kochi</option>
<option value="Other">Other</option></select>
<span class="error" id="cityError"></span>
</div>
<!-- Terms and Conditions -->
<div class="terms-group">
<label class="checkbox-label">
<input
type="checkbox"
id="terms"
name="terms"
required>
<span>
I agree to the Terms & Conditions
</span>
</label>
<span class="error" id="termsError"></span>
</div>
<!-- Buttons -->
<div class="button-group">
<button type="submit" id="registerButton">
Register
</button>
<button type="reset" id="resetButton">
Reset
</button>
</div>
</form>
<!-- Success Message -->
<div
id="successMessage"
class="success-message"
aria-live="polite">
</div>
</div>
<div class="footer">
Client-Side Validation • HTML5 • CSS3 • JavaScript
</div>
</div>
<!-- External JavaScript -->
<script src="script.js"></script>
</body>
</html>
```

STYLE.CSS:

```
* {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}
body {
    font-family: Arial, Helvetica, sans-serif;
    background-color: #f4f1ea;
    color: #111111;
    min-height: 100vh;
    padding: 30px 15px;
}
.container {
    width: 100%;
    max-width: 850px;
    margin: auto;
}
.header {
    background-color: #111111;
    color: #ffffff;
    padding: 22px;
    display: flex;
    align-items: center;
    gap: 15px;
    border-radius: 12px 12px 0 0;
}
.brand-box {
    width: 50px;
    height: 50px;
    background-color: #f4c430;
    color: #111111;
    display: flex;
    align-items: center;
    justify-content: center;
    font-size: 28px;
    font-weight: bold;
    border: 3px solid #ffffff;}
.small-title {
    font-size: 12px;
    font-weight: bold;
    letter-spacing: 2px;
    color: #f4c430;}
.header h1 {
    font-size: 25px;
    margin-top: 4px;}
.status {
    margin-left: auto;
    padding: 8px 12px;
    background-color: #d62828;
    color: #ffffff;
    font-size: 11px;
```

```
    font-weight: bold;
    border-radius: 20px;}
.intro {
    background-color: #ffffff;
    padding: 18px 22px;
    border-left: 4px solid #f4c430;
    border-right: 4px solid #f4c430;}
.intro p {
    color: #555555;
    line-height: 1.6;}
.progress-section {
    background-color: #ffffff;
    padding: 15px 22px;
    border-left: 4px solid #f4c430;
    border-right: 4px solid #f4c430;}
.progress-info {
    display: flex;
    justify-content: space-between;
    font-size: 13px;
    font-weight: bold;
    margin-bottom: 7px;}
.progress-bar {
    width: 100%;
    height: 10px;
    background-color: #dddddd;
    border: 1px solid #111111;}
#progressFill {
    height: 100%;
    width: 0%;
    background-color: #16803c;
    transition: width 0.3s ease;}
.form-card {
    background-color: #ffffff;
    padding: 30px;
    border: 3px solid #111111;
    border-radius: 0 0 12px 12px;
    box-shadow: 8px 8px 0 #111111;}
.form-card h2 {
    font-size: 24px;
    margin-bottom: 25px;
    border-bottom: 3px solid #f4c430;
    padding-bottom: 10px;}
.form-group {
    margin-bottom: 20px;}
.form-group > label {
    display: block;
    font-weight: bold;
    margin-bottom: 7px;}
input[type="text"],
input[type="email"],
input[type="password"],
```

```
input[type="date"],
select {
    width: 100%;
    padding: 12px;
    border: 2px solid #222222;
    border-radius: 6px;
    font-size: 15px;
    background-color: #ffffff;
    outline: none;}
input:focus,
select:focus {
    border-color: #f4c430;
    box-shadow: 0 0 0 3px #f4c430;}
.invalid {
    border-color: #d62828 !important;
    background-color: #fff3f3 !important;}
.valid {
    border-color: #16803c !important;
    background-color: #f1fff5 !important;}
.error {
    display: block;
    min-height: 18px;
    margin-top: 5px;
    color: #d62828;
    font-size: 13px;
    font-weight: bold;
}
.radio-group {
    display: flex;
    gap: 20px;
    flex-wrap: wrap;
    padding: 10px 0;}
.radio-option {
    display: flex;
    align-items: center;
    gap: 7px;
    cursor: pointer;}
.radio-option input {
    accent-color: #111111;
    width: 17px;
    height: 17px;}
.password-info {
    display: flex;
    justify-content: space-between;
    margin-top: 6px;
    font-size: 12px;
    color: #666666;}
.terms-group {
    margin-top: 10px;
    margin-bottom: 20px;}
```

```
.checkbox-label {
    display: flex;
    align-items: center;
    gap: 9px;
    cursor: pointer;
    font-size: 14px;
    font-weight: bold;}
.checkbox-label input {
    width: 18px;
    height: 18px;
    accent-color: #111111;}
.button-group {
    display: flex;
    gap: 12px;
    margin-top: 25px;}
button {
    padding: 13px 25px;
    border: 2px solid #111111;
    border-radius: 6px;
    font-size: 15px;
    font-weight: bold;
    cursor: pointer;
    transition: 0.2s;}
#registerButton {
    background-color: #111111;
    color: #ffffff;}
#resetButton {
    background-color: #f4c430;
    color: #111111;}
button:hover {
    transform: translateY(-2px);
    box-shadow: 4px 4px 0 #111111;}
button:active {
    transform: translateY(0);
    box-shadow: none;}
.success-message {
    display: none;
    margin-top: 25px;
    padding: 20px;
    background-color: #e9f8ee;
    border: 3px solid #16803c;
    color: #145c2e;
    line-height: 1.7;}
.success-message.show {
    display: block;}
.footer {
    text-align: center;
    margin-top: 20px;
    font-size: 12px;
    color: #666666;}
```

```
@media (max-width: 600px) {
   body {
      padding: 15px 10px;}
   .header {
      flex-wrap: wrap;}
   .status {
      margin-left: 0;
      width: 100%;
      text-align: center;}
   .form-card {
      padding: 20px;
      box-shadow: 5px 5px 0 #111111;
   }
   .radio-group {
      flex-direction: column;
      gap: 10px;}
   .button-group {
      flex-direction: column; }
   button {
      width: 100%;
   }
}
```

JAVASCRIPT — validation.js:

```
const registrationForm=document.getElementById("registrationForm");
const fullName=document.getElementById("fullName");
const email=document.getElementById("email");
const mobile=document.getElementById("mobile");
const password=document.getElementById("password");
const confirmPassword=document.getElementById("confirmPassword");
const dob=document.getElementById("dob");
const city=document.getElementById("city");
const terms=document.getElementById("terms");
const nameError=document.getElementById("nameError");
const emailError=document.getElementById("emailError");
const mobileError=document.getElementById("mobileError");
const passwordError=document.getElementById("passwordError");
const confirmPasswordError=document.getElementById("confirmPasswordError");
const genderError=document.getElementById("genderError");
const dobError=document.getElementById("dobError");
const cityError=document.getElementById("cityError");
const termsError=document.getElementById("termsError");
const successMessage=document.getElementById("successMessage");
const statusText=document.getElementById("statusText");
const progressText=document.getElementById("progressText");
const progressFill=document.getElementById("progressFill");
const passwordCount=document.getElementById("passwordCount");
const passwordStrength=document.getElementById("passwordStrength");
const today=new Date();
const todayString=today.toISOString().split("T")[0];
dob.max=todayString;
```

```
const nameRegex=/^[A-Za-z ]+$/;
const emailRegex=/^[^\s@]+@[^\s@]+\.[^\s@]+$/;
const mobileRegex=/^\d{10}$/;
const passwordRegex=/^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@$!%*?&])[A-Za-z\d@$!%*?&]{8,}$/;
function showError(inputElement,errorElement,message){
errorElement.textContent=message;
inputElement.classList.add("invalid");
inputElement.classList.remove("valid");
}
function clearError(inputElement,errorElement){
errorElement.textContent="";
inputElement.classList.remove("invalid");
}
function markValid(inputElement){
inputElement.classList.remove("invalid");
inputElement.classList.add("valid");
}
function clearAllErrors(){
const errorElements=document.querySelectorAll(".error");
errorElements.forEach(function(error){
error.textContent="";
});
const inputs=document.querySelectorAll("input,select");
inputs.forEach(function(input){
input.classList.remove("invalid");
input.classList.remove("valid");
});
}
function updateProgress(){
let completedFields=0;
const totalFields=9;
if(fullName.value.trim()!==""&&nameRegex.test(fullName.value.trim())) completedFields++;
if(email.value.trim()!==""&&emailRegex.test(email.value.trim())) completedFields++;
if(mobile.value.trim()!==""&&mobileRegex.test(mobile.value.trim())) completedFields++;
if(password.value.trim()!==""&&passwordRegex.test(password.value.trim())) completedFields++;
if(confirmPassword.value.trim()!==""&&confirmPassword.value===password.value) completedFields++;
const selectedGender=document.querySelector('input[name="gender"]:checked');
if(selectedGender) completedFields++;
if(dob.value!==""){
const birthDate=new Date(dob.value);
const currentDate=new Date();
let age=currentDate.getFullYear()-birthDate.getFullYear();
const monthDifference=currentDate.getMonth()-birthDate.getMonth();
if(monthDifference<0||(monthDifference===0&&currentDate.getDate()<birthDate.getDate())) age--;
if(age>=18) completedFields++;
}
if(city.value!=="") completedFields++;
if(terms.checked) completedFields++;
const percentage=Math.round((completedFields/totalFields)*100);
progressText.textContent=percentage+"%";
progressFill.style.width=percentage+"%";
if(percentage===100){
```

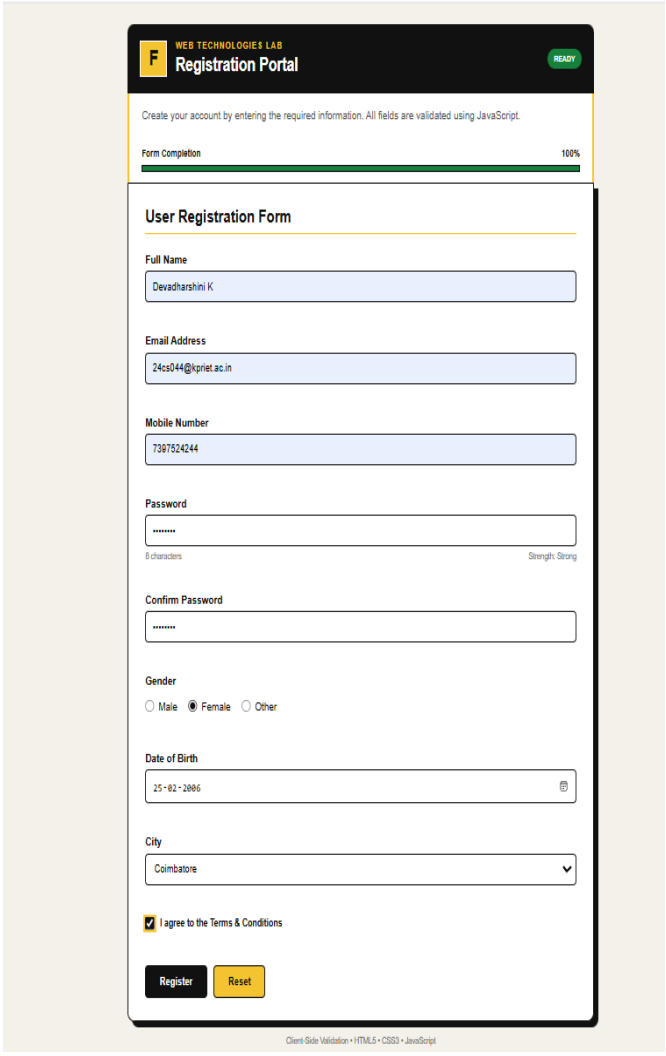
```
statusText.textContent="READY";
statusText.style.backgroundColor="#16803c";
}else{
statusText.textContent="INCOMPLETE";
statusText.style.backgroundColor="#d62828";
}
}
function updatePasswordStrength(){
const passwordValue=password.value;
passwordCount.textContent=passwordValue.length+" characters";
if(passwordValue.length===0){
passwordStrength.textContent="Strength: -";
}else if(passwordValue.length<8){
passwordStrength.textContent="Strength: Too Short";
}else if(passwordRegex.test(passwordValue)){
passwordStrength.textContent="Strength: Strong";
}else{
passwordStrength.textContent="Strength: Weak";
}
}
function validateForm(event){
event.preventDefault();
clearAllErrors();
successMessage.classList.remove("show");
successMessage.innerHTML="";
let isValid=true;
const nameValue=fullName.value.trim();
const emailValue=email.value.trim();
const mobileValue=mobile.value.trim();
const passwordValue=password.value.trim();
const confirmPasswordValue=confirmPassword.value.trim();
const dobValue=dob.value.trim();
const cityValue=city.value.trim();
if(nameValue===""){
showError(fullName,nameError,"Full Name is required.");
isValid=false;
}else if(nameValue.length<3){
showError(fullName,nameError,"Name must contain at least 3 characters.");
isValid=false;
}else if(!nameRegex.test(nameValue)){
showError(fullName,nameError,"Name can contain only alphabets and spaces.");
isValid=false;
}else{
markValid(fullName);
}
if(emailValue===""){
showError(email,emailError,"Email Address is required.");
isValid=false;
}else if(!emailRegex.test(emailValue)){
showError(email,emailError,"Enter a valid email address.");
isValid=false;
}else{
```

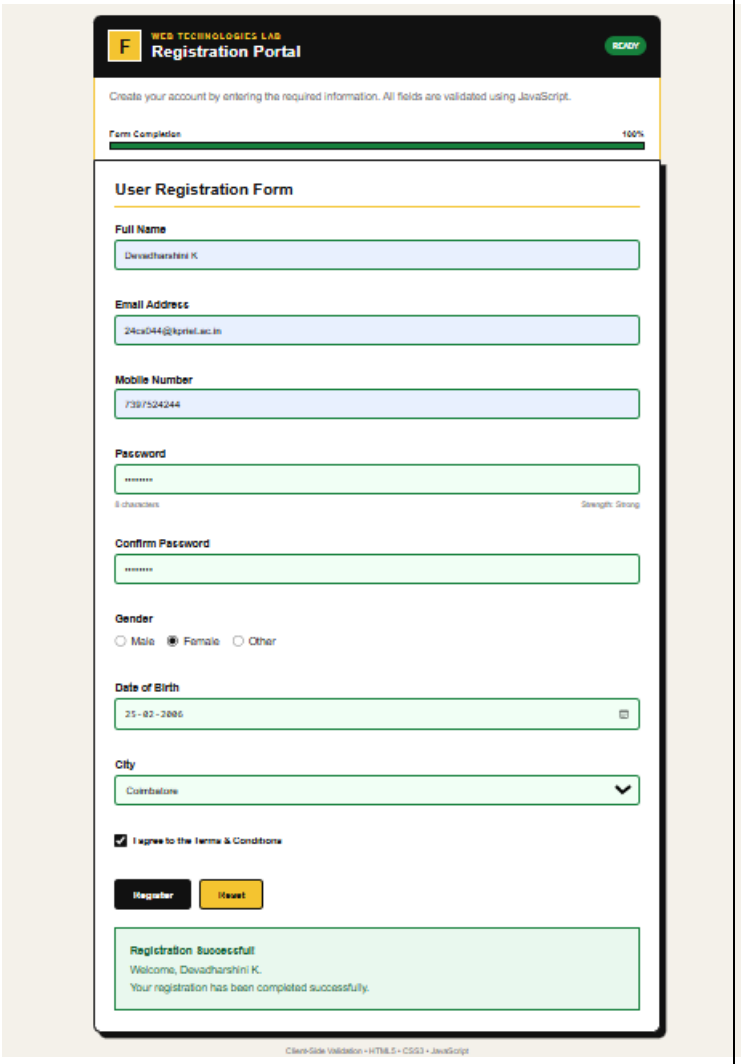
```
markValid(email);
}
if(mobileValue===""){
showError(mobile,mobileError,"Mobile Number is required.");
isValid=false;
}else if(!mobileRegex.test(mobileValue)){
showError(mobile,mobileError,"Mobile number must contain exactly 10 digits.");
isValid=false;
}else{
markValid(mobile);
}
if(passwordValue===""){
showError(password,passwordError,"Password is required.");
isValid=false;
}else if(passwordValue.length<8){
showError(password,passwordError,"Password must contain at least 8 characters.");
isValid=false;
}else if(!passwordRegex.test(passwordValue)){
showError(password,passwordError,"Password must contain uppercase, lowercase, digit and special character.");
isValid=false;
}else{
markValid(password);
}
if(confirmPasswordValue===""){
showError(confirmPassword,confirmPasswordError,"Please confirm your password.");
isValid=false;
}else if(confirmPasswordValue!==passwordValue){
showError(confirmPassword,confirmPasswordError,"Passwords do not match.");
isValid=false;
}else{
markValid(confirmPassword);
}
const selectedGender=document.querySelector('input[name="gender"]:checked');
if(!selectedGender){
genderError.textContent="Please select your gender.";
isValid=false;
}
if(dobValue===""){
dobError.textContent="Date of Birth is required.";
dob.classList.add("invalid");
isValid=false;
}else{
const birthDate=new Date(dobValue);
const currentDate=new Date();
let age=currentDate.getFullYear()-birthDate.getFullYear();
const monthDifference=currentDate.getMonth()-birthDate.getMonth();
if(monthDifference<0||(monthDifference===0&&currentDate.getDate()<birthDate.getDate())) age--;
if(birthDate>currentDate){
dobError.textContent="Date of Birth cannot be in the future.";
dob.classList.add("invalid");
isValid=false;
}else if(age<18){
```

```
dobError.textContent="You must be at least 18 years old.";
dob.classList.add("invalid");
isValid=false;
}else{
markValid(dob);
}
}
if(cityValue===""){
showError(city,cityError,"Please select your city.");
isValid=false;
}else{
markValid(city);
}
if(!terms.checked){
termsError.textContent="You must agree to the Terms & Conditions.";
isValid=false;
}
if(isValid){
const selectedName=fullName.value.trim();
successMessage.innerHTML=`<strong>Registration Successful!</strong><br>Welcome,
${selectedName}.<br>Your registration has been completed successfully.`;
successMessage.classList.add("show");
statusText.textContent="SUCCESS";
statusText.style.backgroundColor="#16803c";
progressText.textContent="100%";
progressFill.style.width="100%";
}else{
statusText.textContent="CHECK ERRORS";
statusText.style.backgroundColor="#d62828";
}
updateProgress();
}
registrationForm.addEventListener("submit",validateForm);
password.addEventListener("input",function(){
updatePasswordStrength();
updateProgress();
});
mobile.addEventListener("input",function(){
mobile.value=mobile.value.replace(/\D/g,"");
if(mobile.value.length>10) mobile.value=mobile.value.substring(0,10);
updateProgress();
});
fullName.addEventListener("input",updateProgress);
email.addEventListener("input",updateProgress);
confirmPassword.addEventListener("input",updateProgress);
dob.addEventListener("change",updateProgress);
city.addEventListener("change",updateProgress);
terms.addEventListener("change",updateProgress);
const genderOptions=document.querySelectorAll('input[name="gender"]');
genderOptions.forEach(function(gender){
gender.addEventListener("change",updateProgress);
});
```

```
registrationForm.addEventListener("reset",function(){
setTimeout(function(){
clearAllErrors();
successMessage.classList.remove("show");
successMessage.innerHTML="";
passwordCount.textContent="0 characters";
passwordStrength.textContent="Strength: -";
statusText.textContent="INCOMPLETE";
statusText.style.backgroundColor="#d62828";
progressText.textContent="0%";
progressFill.style.width="0%";
},0);
});
updateProgress();
```

OUTPUT:





RUBRICS:

Rubrics	Actual Marks	Awarded Marks
Understanding (R1)	4	
Timeline (R2)	4	
Design (R3)	4	
Development (R4)	4	
Innovation (R5)	4	
Total Marks	**20**	

RESULT:

Thus, the User Registration Form was developed successfully with client-side validation using JavaScript. The form validates the user inputs and displays appropriate error messages for invalid data.

Ex No: 04	Implementing web applications with API calls using Java Script

AIM:

To develop a web application that retrieves data from a public REST API using JavaScript Fetch API, processes JSON data, and displays the API response dynamically using DOM manipulation.

PROCEDURE:

1. Create a project folder named API_Web_Application.
2. Create three files named index.html, style.css, and api.js.
3. Create the HTML structure with a container and add the heading **Public API Data Viewer**.
4. Add a Load Data button and sections for displaying loading messages, error messages, success messages, and API results.
5. Apply CSS styling to the page, container, button, loading message, error message, and result cards.
6. Create the loadData() function in JavaScript and use the Fetch API to request data from a public REST API.
7. Use async and await to handle the API request and convert the response into JSON using response.json().
8. Process the JSON data using a loop and extract the user name, email, phone, and website details.
9. Dynamically create result cards using createElement(), innerHTML, and appendChild().
10. Display loading and success messages and handle API or network errors using try...catch.

HTML USED:

- <!DOCTYPE html>
- <html>
- <head>
- <title>
- <body>
-
- <h2>
- <button>
- <p>

CSS PROPERTIES USED:

- margin
- padding
- background-color
- font-family
- text-align
- width
- max-width
- border
- border-radius
- box-shadow
- display

- grid
- gap
- color
- font-size
- font-weight
- cursor
- box-sizing

JAVASCRIPT CONCEPTS USED:

- Variables
- Functions
- Arrays
- Objects
- Fetch API
- REST API
- Asynchronous Programming
- Promises
- async
- await
- JSON
- response.json()
- try...catch
- Event Handling
- DOM Manipulation
- document.getElementById()
- createElement()
- appendChild()
- innerHTML
- textContent
- forEach()

INDEX.HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width,initial-scale=1.0">
<title>API Web Application</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
<header class="header">
<div class="logo">API</div>
<div>
<p class="small-title">WEB TECHNOLOGIES LAB</p>
```

```
<h1>Public API Data Viewer</h1>
</div>
</header>
<section class="intro">
<p>Fetch and explore user information from a public REST API using JavaScript.</p>
</section>
<section class="controls">
<label for="searchInput">Search User</label>
<input type="text" id="searchInput" placeholder="Enter user name...">
<button id="loadButton">Load Data</button>
</section>
<section class="status-area">
<div id="loadingMessage" class="loading">Loading data...</div>
<div id="errorMessage" class="error"></div>
<div id="successMessage" class="success"></div>
</section>
<section class="result-info">
<h3 id="recordCount">Records: 0</h3>
</section>
<div id="resultsContainer" class="results-container"></div>
</div>
<script src="api.js"></script>
</body>
</html>
```

STYLE.CSS:

```
* {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}
body {
    font-family: Arial, Helvetica, sans-serif;
    background-color: #f3f0e8;
    color: #111111;
    min-height: 100vh;
    padding: 30px 15px;
}
.container {
    width: 100%;
    max-width: 1000px;
    margin: auto;
}
.header {
    background-color: #111111;
    color: white;
    padding: 25px;
    display: flex;
    align-items: center;
    gap: 18px;
    border-radius: 12px 12px 0 0;}
```

```
.logo {
    width: 60px;
    height: 60px;
    display: flex;
    align-items: center;
    justify-content: center;
    background-color: #f4c430;
    color: #111111;
    font-size: 22px;
    font-weight: bold;
    border: 3px solid white;
}
.small-title {
    color: #f4c430;
    font-size: 12px;
    font-weight: bold;
    letter-spacing: 2px;
    margin-bottom: 5px;
}
.header h1 {
    font-size: 27px;
}
.intro {
    background-color: white;
    padding: 20px 25px;
    border-left: 4px solid #f4c430;
    border-right: 4px solid #f4c430;
}
.intro p {
    color: #555555;
    line-height: 1.6;
}
.controls {
    background-color: white;
    padding: 25px;
    display: flex;
    align-items: center;
    gap: 12px;
    border-left: 4px solid #f4c430;
    border-right: 4px solid #f4c430;
    flex-wrap: wrap;
}
.controls label {
    font-weight: bold;
    width: 100%;
}
.controls input {
    flex: 1;
    min-width: 200px;
    padding: 13px;
    border: 2px solid #111111;
    border-radius: 6px;
```

```
font-size: 15px;
outline: none;}
.controls input:focus {
    border-color: #f4c430;
    box-shadow: 0 0 0 3px #f4c430;
}
button {
    padding: 13px 25px;
    background-color: #111111;
    color: white;
    border: 2px solid #111111;
    border-radius: 6px;
    font-weight: bold;
    font-size: 15px;
    cursor: pointer;
    transition: 0.2s;
}
button:hover {
    background-color: #f4c430;
    color: #111111;
    transform: translateY(-2px);
    box-shadow: 4px 4px 0 #111111;
}
button:active {
    transform: translateY(0);
    box-shadow: none;
}
.status-area {
    background-color: white;
    padding: 5px 25px;
}
.loading {
    display: none;
    padding: 15px;
    text-align: center;
    font-weight: bold;
    color: #111111;
    background-color: #fff4bf;
    border: 2px solid #f4c430;
}
.error {
    display: none;
    padding: 15px;
    color: #a80000;
    background-color: #ffe6e6;
    border: 2px solid #d62828;
    font-weight: bold;
}
.success {
    display: none;
    padding: 15px;
    color: #145c2e;
```

```
    background-color: #e8f7ed;
    border: 2px solid #16803c;
    font-weight: bold;
}
.result-info {
    background-color: white;
    padding: 20px 25px;
    border-left: 4px solid #f4c430;
    border-right: 4px solid #f4c430;
}
.result-info h3 {
    font-size: 17px;
}
.results-container {
    background-color: white;
    padding: 10px 25px 35px;
    display: grid;
    grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));
    gap: 20px;
    border-radius: 0 0 12px 12px;
    border-left: 4px solid #f4c430;
    border-right: 4px solid #f4c430;
    border-bottom: 3px solid #111111;
    box-shadow: 8px 8px 0 #111111;
}
.user-card {
    background-color: #ffffff;
    border: 2px solid #111111;
    border-radius: 8px;
    padding: 20px;
    transition: 0.2s;
}
.user-card:hover {
    transform: translateY(-5px);
    box-shadow: 6px 6px 0 #111111;
}
.user-number {
    display: inline-block;
    background-color: #f4c430;
    color: #111111;
    padding: 5px 10px;
    font-size: 12px;
    font-weight: bold;
    border-radius: 20px;
    margin-bottom: 12px;
}
.user-card h2 {
    font-size: 20px;
    margin-bottom: 15px;
}
.user-detail {
    margin-bottom: 10px;
```

```
font-size: 14px;
line-height: 1.5;}
.user-detail strong {
    display: block;
    color: #555555;
    font-size: 12px;
    text-transform: uppercase;
    margin-bottom: 2px;
}
.user-detail a {
    color: #111111;
    font-weight: bold;
    word-break: break-word;
}
.no-results {
    grid-column: 1 / -1;
    text-align: center;
    padding: 30px;
    color: #666666;
    font-weight: bold;
}
@media (max-width: 600px) {
    body {
        padding: 15px 10px;
    }
    .header {
        flex-direction: column;
        align-items: flex-start;
    }
    .controls {
        flex-direction: column;
        align-items: stretch;
    }
    .controls input {
        width: 100%;
    }
    .controls button {
        width: 100%;
    }
    .results-container {
        grid-template-columns: 1fr;
        box-shadow: 5px 5px 0 #111111;
    }
}
```

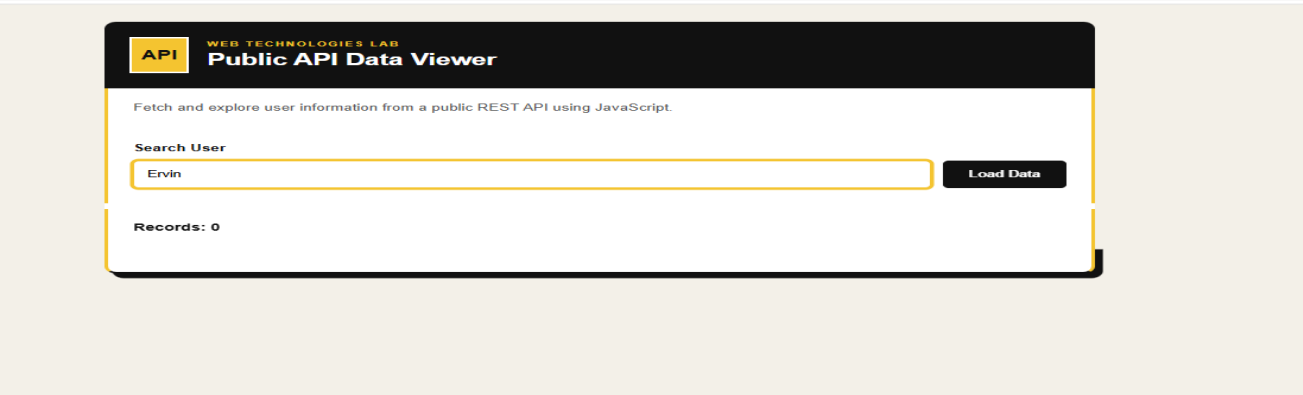
API.JX:

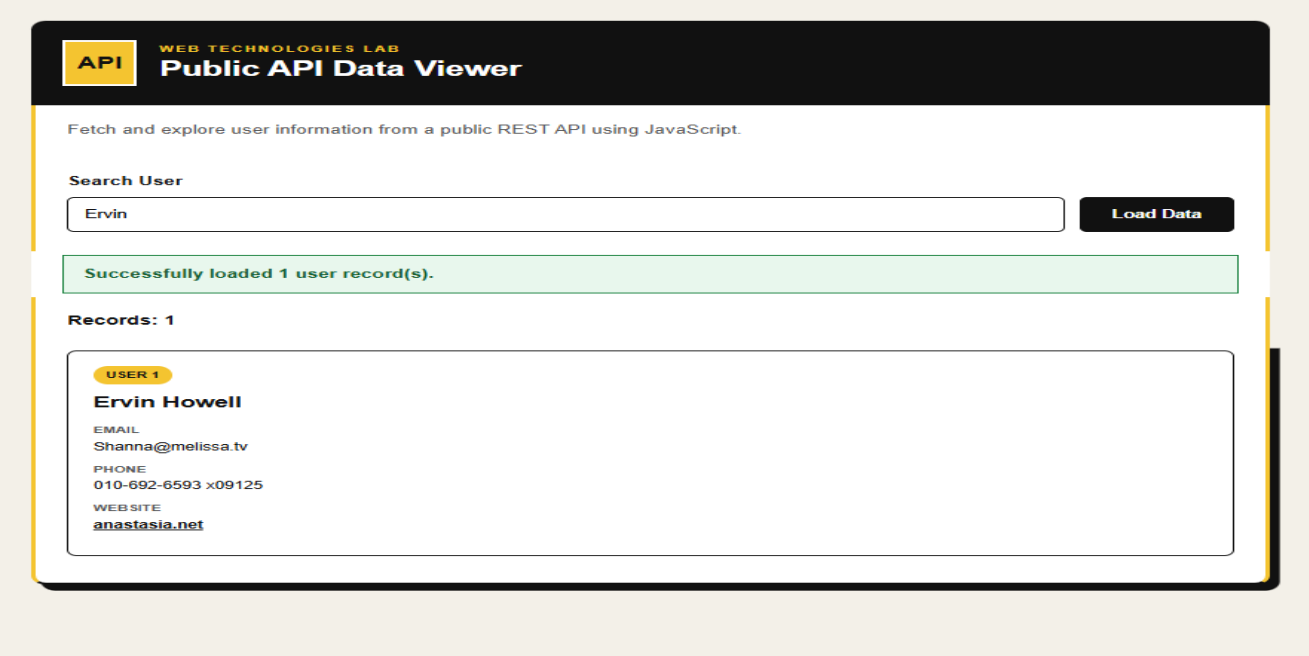
```
const loadButton=document.getElementById("loadButton");
const searchInput=document.getElementById("searchInput");
const loadingMessage=document.getElementById("loadingMessage");
const errorMessage=document.getElementById("errorMessage");
const successMessage=document.getElementById("successMessage");
const resultsContainer=document.getElementById("resultsContainer");
const recordCount=document.getElementById("recordCount");
```

```
const API_URL="https://jsonplaceholder.typicode.com/users";
async function loadData(){
errorMessage.style.display="none";
successMessage.style.display="none";
resultsContainer.innerHTML="";
recordCount.textContent="Records: 0";
loadingMessage.style.display="block";
try{
const response=await fetch(API_URL);
if(!response.ok){
throw new Error("Unable to fetch data. Server returned an error.");
}
const users=await response.json();
const searchKeyword=searchInput.value.trim().toLowerCase();
const filteredUsers=users.filter(function(user){
return user.name.toLowerCase().includes(searchKeyword);
});
loadingMessage.style.display="none";
if(filteredUsers.length===0){
resultsContainer.innerHTML=`<div class="no-results">No users found for "${searchKeyword}".</div>`;
recordCount.textContent="Records: 0";
return;
}
filteredUsers.forEach(function(user,index){
createUserCard(user,index+1);
});
recordCount.textContent="Records: "+filteredUsers.length;
successMessage.textContent=`Successfully loaded ${filteredUsers.length} user record(s).`;
successMessage.style.display="block";
}catch(error){
loadingMessage.style.display="none";
errorMessage.textContent="Error: Unable to load API data. Please check your internet connection and
try again.";
errorMessage.style.display="block";
console.error("API Error:",error);
}
}
function createUserCard(user,number){
const card=document.createElement("div");
card.className="user-card";
const userNumber=document.createElement("span");
userNumber.className="user-number";
userNumber.textContent="USER "+number;
const name=document.createElement("h2");
name.textContent=user.name;
const email=document.createElement("div");
email.className="user-detail";
email.innerHTML=`<strong>Email</strong>${user.email}`;
const phone=document.createElement("div");
phone.className="user-detail";
phone.innerHTML=`<strong>Phone</strong>${user.phone}`;
const website=document.createElement("div");
```

```
website.className="user-detail";
website.innerHTML=`<strong>Website</strong><ahref="https://${user.website}"
target="_blank">${user.website}</a>`;
card.appendChild(userNumber);
card.appendChild(name);
card.appendChild(email);
card.appendChild(phone);
card.appendChild(website);
resultsContainer.appendChild(card);
}
loadButton.addEventListener("click",loadData);
searchInput.addEventListener("keydown",function(event){
if(event.key==="Enter"){
loadData();
}
});
```

OUTPUT:





RUBRICKS:

Rubrics	Actual Marks	Awarded Marks
Understanding (R1)	4	
Timeline (R2)	4	
Design (R3)	4	
Development (R4)	4	
Innovation (R5)	4	
Total Marks	**20**	

RESULT:

Thus, the API-based web application was developed successfully using JavaScript Fetch API. The data was fetched from the API, processed as JSON, and displayed dynamically on the webpage.